

Data modeling: CSCI E-106

Applied Linear Statistical Models
Chapter 7 – Multiple Regression II

Illustration for the extra sum of squares

Example 1: A study of the relation of amount of body fat: a sample of 20 healthy females: 25-34 years old

- Y : body fat
- X_1 : triceps skinfold thickness
- X_2 : thigh circumference
- X_3 : midarm circumference

It would be very helpful if a regression model with some or all these predictor variables could provide reliable estimates of amount of body fat.

Illustration for the extra sum of squares, cont'd

Table : Basic Data-Body Fat Example.

Subject	Triceps Skinfold Thickness	Thigh Circumference	Midarm Circumference	Body Fat
i	X_{i1}	X_{i2}	X_{i3}	Y_i
1	19.50	43.10	29.10	11.90
2	24.70	49.80	28.20	22.80
3	30.70	51.90	37.00	18.70
4	29.80	54.30	31.10	20.10
5	19.10	42.20	30.90	12.90
6	25.60	53.90	23.70	21.70
7	31.40	58.50	27.60	27.10
8	27.90	52.10	30.60	25.40
9	22.10	49.90	23.20	21.30
10	25.50	53.50	24.80	19.30
11	31.10	56.60	30.00	25.40
12	30.40	56.70	28.30	27.20
13	18.70	46.50	23.00	11.70
14	19.70	44.20	28.60	17.80
15	14.60	42.70	21.30	12.80
16	29.50	54.40	30.10	23.90
17	27.70	55.30	25.70	22.60
18	30.20	58.60	24.60	25.40
19	22.70	48.20	27.10	14.80
20	25.20	51.00	27.50	21.10

Illustration for the extra sum of squares, cont'd

(a) Regression of Y on X_1

(b) Regression of Y on X_2

TABLE 7.2
Regression
Results for
Several Fitted
Models—Body
Fat Example.

(a) Regression of Y on X_1 $\hat{Y} = -1.496 + .8572X_1$			
Source of Variation	SS	df	MS
Regression	352.27	1	352.27
Error	143.12	18	7.95
Total	495.39	19	
Variable	Estimated Regression Coefficient	Estimated Standard Deviation	t^*
X_1	$b_1 = .8572$	$s\{b_1\} = .1288$	6.66
(b) Regression of Y on X_2 $\hat{Y} = -23.634 + .8565X_2$			
Source of Variation	SS	df	MS
Regression	381.97	1	381.97
Error	113.42	18	6.30
Total	495.39	19	
Variable	Estimated Regression Coefficient	Estimated Standard Deviation	t^*
X_2	$b_2 = .8565$	$s\{b_2\} = .1100$	7.79

Illustration for the extra sum of squares, cont'd

(c) Regression of Y on X_1 and X_2
 X_3

(d) Regression of Y on X_1 , X_2 and X_3

TABLE 7.2
(Continued).

(c) Regression of Y on X_1 and X_2 $\hat{Y} = -19.174 + .2224X_1 + .6594X_2$			
Source of Variation	SS	df	MS
Regression	385.44	2	192.72
Error	109.95	17	6.47
Total	495.39	19	
Variable	Estimated Regression Coefficient	Estimated Standard Deviation	t^*
X_1	$b_1 = .2224$	$s\{b_1\} = .3034$	.73
X_2	$b_2 = .6594$	$s\{b_2\} = .2912$	2.26
(d) Regression of Y on X_1 , X_2 , and X_3 $\hat{Y} = 117.08 + 4.334X_1 - 2.857X_2 - 2.186X_3$			
Source of Variation	SS	df	MS
Regression	396.98	3	132.33
Error	98.41	16	6.15
Total	495.39	19	
Variable	Estimated Regression Coefficient	Estimated Standard Deviation	t^*
X_1	$b_1 = 4.334$	$s\{b_1\} = 3.016$	1.44
X_2	$b_2 = -2.857$	$s\{b_2\} = 2.582$	-1.11
X_3	$b_3 = -2.186$	$s\{b_3\} = 1.596$	-1.37

Illustration for the extra sum of squares, cont'd

Notations:

- Assume X_1 is in the model
 - $SSR(X_1)$: The regression sum of squares
 - $SSE(X_1)$: The error sum of squares
 - measure the marginal effect of adding X_2 (another variable) to the regression model when X_1 is already in the model
 - $SSR(X_2|X_1)$: The extra sum of squares gained by adding X_2
- Assume X_1 and X_2 are in the model
 - $SSR(X_1, X_2)$: The regression sum of squares
 - $SSE(X_1, X_2)$: The error sum of squares
 - measure the marginal effect of adding X_3 (another variable) to the regression model when X_1 and X_2 are already in the model
 - $SSR(X_3|X_1, X_2)$: The extra sum of squares gained by adding X_3

Illustration for the extra sum of squares, cont'd

(a) Regression of Y on X_1
 $\hat{Y} = -1.496 + .8572X_1$

Source of Variation	SS	df	MS
Regression	352.27	1	352.27
Error	143.12	18	7.95
Total	495.39	19	

Variable	Estimated Regression Coefficient	Estimated Standard Deviation	t^*
X_1	$b_1 = .8572$	$s\{b_1\} = .1288$	6.66

(c) Regression of Y on X_1 and X_2
 $\hat{Y} = -19.174 + .2224X_1 + .6594X_2$

Source of Variation	SS	df	MS
Regression	385.44	2	192.72
Error	109.95	17	6.47
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Variable	Estimated Regression Coefficient	Estimated Standard Deviation	t^*
X_1	$b_1 = .2224$	$s\{b_1\} = .3034$	.73
X_2	$b_2 = .6594$	$s\{b_2\} = .2912$	2.26

An extra sum of squares:

$$SSR(X_2 | X_1) = SSE(X_1) - SSE(X_1, X_2) = 143.12 - 109.95 = 33.17$$

$$= SSR(X_1, X_2) - SSR(X_1) = 385.44 - 352.27 = 33.17$$

$SSR(X_2 | X_1)$

- the marginal increase in the regression sum of squares (SSR)
- reflects the additional or extra reduction in the error sum of squares (SSE) associated with X_2 , given that X_1 is already included in the model

Illustration for the extra sum of squares, cont'd

- The marginal reduction in the $SSE = \text{The marginal increase in } SSR$
- $SSTO = SSR + SSE$:
 - measure the variability of Y_i and does not depend on the regression model fitted
 - Any reduction in SSE implies an identical increase in SSR

Illustration for the extra sum of squares, cont'd

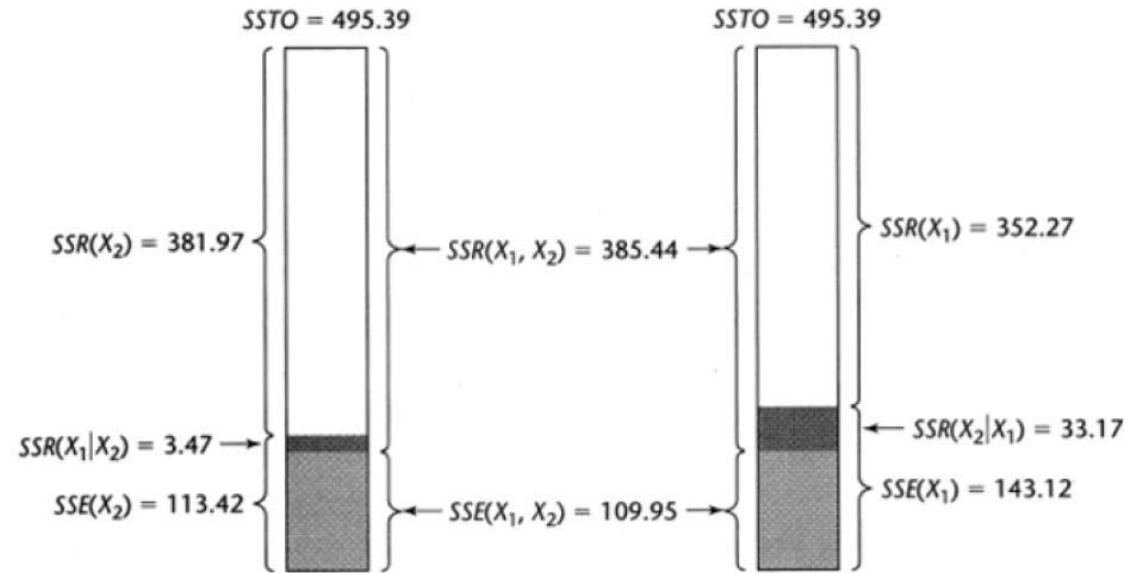


Figure : Schematic Representation of Extra Sums of Squares-Body Fat Example.

$$\begin{aligned} SSR(X_2|X_1) &= SSE(X_1) - SSE(X_1, X_2) = 143.12 - 109.95 = 33.17 \\ &= SSR(X_1, X_2) - SSR(X_1) = 385.44 - 352.27 = 33.17 \end{aligned}$$

Illustration for the extra sum of squares, cont'd

(c) Regression of Y on X_1 and X_2
 $\hat{Y} = -19.174 + .2224X_1 + .6594X_2$

Source of Variation	SS	df	MS
Regression	385.44	2	192.72
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Variable	Estimated Regression Coefficient	Estimated Standard Deviation	t^*
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X_2	$b_2 = .6594$	$s\{b_2\} = .2912$	2.26

(d) Regression of Y on X_1 , X_2 , and X_3
 $\hat{Y} = 117.08 + 4.334X_1 - 2.857X_2 - 2.186X_3$

Source of Variation	SS	df	MS
Regression	396.98	3	132.33
Error	98.41	16	6.15
Total	495.39	19	

Variable	Estimated Regression Coefficient	Estimated Standard Deviation	t^*
X_1	$b_1 = 4.334$	$s\{b_1\} = 3.016$	1.44
X_2	$b_2 = -2.857$	$s\{b_2\} = 2.582$	-1.11
X_3	$b_3 = -2.186$	$s\{b_3\} = 1.596$	-1.37

An extra sum of squares: adding X_3

$$\begin{aligned}
 SSR(X_3 | X_1, X_2) &= SSE(X_1, X_2) - SSE(X_1, X_2, X_3) = 109.95 - 98.41 = 11.54 \\
 &= SSR(X_1, X_2, X_3) - SSR(X_1, X_2) = 396.98 - 385.44 = 11.54
 \end{aligned}$$

An extra sum of squares: adding X_2, X_3

$$\begin{aligned}
 SSR(X_2, X_3 | X_1) &= SSE(X_1) - SSE(X_1, X_2, X_3) = 143.12 - 98.41 = 44.71 \\
 &= SSR(X_1, X_2, X_3) - SSR(X_1) = 396.98 - 352.27 = 44.71
 \end{aligned}$$

ANOVA Table Containing Decomposition of SSR

		SST	SSR		SSE	
Model 1	Y=X1	495.39	SSR(X1)	352.27	SSE(X1)	143.12
Model 2	Y=X2	495.39	SSR(X2)	381.97	SSEX2)	113.42
Model 3	Y=X3	495.39	SSR(X3)	10.05	SSE(X3)	485.34
Model 4	Y=X1+X2	495.39	SSR(X1,X2)	385.44	SSE(X1,X2)	109.95
Model 5	Y=X1+X2+X3	495.39	SSR(X1,X2,X3)	396.98	SSE(X1,X2,X3)	98.41

	Source of Variation	SSR	
X1	X1	SSR(X1)	352.27
X2	X2/X1	SSR(X1,X2)=SSR(X1,X2)-SSR(X1)	385.44-352.27=33.17
X3	X3/X1,x2	SSR(X3/X1,X2)=SSR(X1,X2,X3)-SSR(X1)	396.98-385.44=11.54

Sum of Squares:

adding X_1

$$SSR(X_1) = 11.54$$

adding X_2

$$SSR(X_2 | X_1) = SSE(X_1) - SSE(X_1, X_2) = 143.12 - 109.95 = 33.17$$

$$= SSR(X_1, X_2) - SSR(X_1) = 385.44 - 352.27 = 33.17$$

adding X_3

$$SSR(X_3 | X_1, X_2) = SSE(X_1, X_2) - SSE(X_1, X_2, X_3) = 109.95 - 98.41 = 11.54$$

$$= SSR(X_1, X_2, X_3) - SSR(X_1, X_2) = 396.98 - 385.44 = 11.54$$

Illustration for the extra sum of squares, cont'd

Extra Sums of Squares

- An extra sum of squares measures the marginal decrease in the error sum of squares when one or several predictor variables are added to the regression model, given that other variables are already in the model.
- Equivalently, one can view the extra sum of squares as measuring the marginal increase in the regression sum of squares
- Extra: $SSE \downarrow$; $SSR \uparrow$

Definitions

Extra Sums of Squares for two variables:

If X_1 is the extra variable:

$$\begin{aligned} SSR(X_1|X_2) &= SSE(X_2) - SSE(X_1, X_2) \\ &= SSR(X_1, X_2) - SSR(X_2) \end{aligned}$$

If X_2 is the extra variable:

$$\begin{aligned} SSR(X_2|X_1) &= SSE(X_1) - SSE(X_1, X_2) \\ &= SSR(X_1, X_2) - SSR(X_1) \end{aligned}$$

ANOVA Table Containing Decomposition of SSR

Example of ANOVA Table With Decomposition Three X Variables.

Source of Variation	<i>SS</i>	<i>df</i>	<i>MS</i>
Regression	$SSR(X_1, X_2, X_3)$	3	$MSR(X_1, X_2, X_3)$
X_1	$SSR(X_1)$	1	$MSR(X_1)$
$X_2 X_1$	$SSR(X_2 X_1)$	1	$MSR(X_2 X_1)$
$X_3 X_1, X_2$	$SSR(X_3 X_1, X_2)$	1	$MSR(X_3 X_1, X_2)$
Error	$SSE(X_1, X_2, X_3)$	$n - 4$	$MSE(X_1, X_2, X_3)$
Total	$SSTO$	$n - 1$	

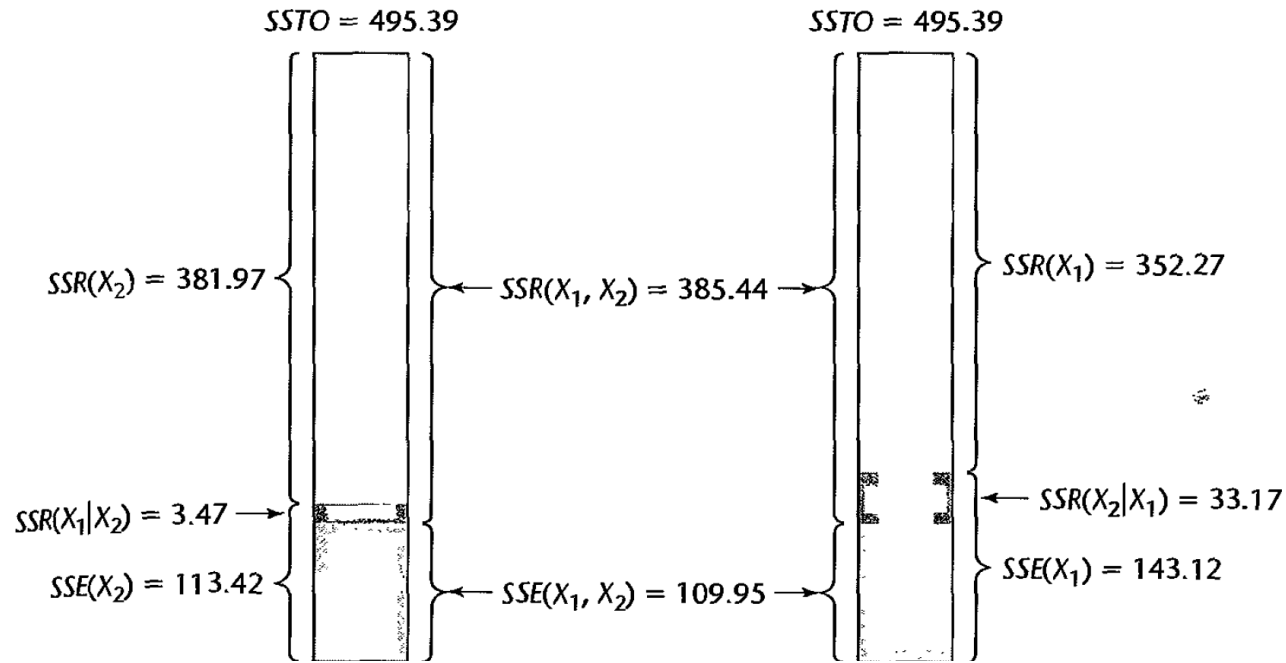
ANOVA Table Containing Decomposition of SSR, cont'd

- Each extra sum of squares involving
 - a single extra X variable has associated with it one degree of freedom
 - two extra X variables have two degrees of freedom
- Mean squares:

$$MSR(X_2|X_1) = \frac{SSR(X_2|X_1)}{1}$$

$$MSR(X_2, X_3|X_1) = \frac{SSR(X_2, X_3|X_1)}{2}$$

ANOVA Table Containing Decomposition of SSR, cont'd



Source of Variation	SS	df	MS
Regression	396.98	3	132.33
X_1	352.27	1	352.27
$X_2 X_1$	33.17	1	33.17
$X_3 X_1, X_2$	11.54	1	11.54
Error	98.41	16	6.15
Total	495.39	19	

Extra sums of squares are of interest because they occur in a variety of tests about regression coefficients where the question of concern is whether certain X variables can be dropped from the regression model.

Test whether a Single $\beta_k = 0$

- Test whether $\beta_k X_k$ can be **dropped** from a multiple regression model

$$H_0 : \beta_k = 0$$

$$H_a : \beta_k \neq 0$$

- Test statistics: $t^* = \frac{b_k}{s\{b_k\}}$
- The general linear test approach: **Full model** vs. **Reduced model**

$$F^* = \frac{SSE(R) - SSE(F)}{df_R - df_F} \div \frac{SSE(F)}{df_F}$$

Test whether a Single $\beta_k = 0$, cont'd

- The general linear test approach (Sec. 2.8) involves an extra sum of squares:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i \quad \text{Full Model}$$

- To test the alternatives:

$$H_0: \beta_3 = 0 \text{ vs. } H_a: \beta_3 \neq 0$$

- When H_0 holds:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i \quad \text{Reduced Model}$$

- The test whether or not $\beta_3 = 0$ is a marginal test, given X_1, X_2 are already in the model

Test whether a Single $\beta_k = 0$, cont'd

- Steps:

1. $SSE(F) = SSE(X_1, X_2, X_3)$, $df_F = n - 4$
2. $SSE(R) = SSE(X_1, X_2)$, $df_R = n - 3$
3. The general linear test statistic (2.70):

$$\begin{aligned} FF^* &= \frac{SSE(R) - SSE(F)}{df_R - df_F} \div \frac{SSE(F)}{df_F} \\ &= \frac{SSE(X_1, X_2) - SSE(X_1, X_2, X_3)}{(n - 3) - (n - 4)} \div \frac{SSE(X_1, X_2, X_3)}{n - 4} \\ &= \frac{SSR(X_3|X_1, X_2)}{1} \div \frac{SSE(X_1, X_2, X_3)}{n - 4} \\ &= \frac{MSR(X_3|X_1, X_2)}{MSE(X_1, X_2, X_3)} \end{aligned}$$

Test whether a Single $\beta_k = 0$, cont'd

TABLE 7.4
ANOVA Table
with
Decomposition
of *SSR*—Body
Fat Example
with Three
Predictor
Variables.

Source of Variation	<i>SS</i>	<i>df</i>	<i>MS</i>
Regression	396.98	3	132.33
X_1	352.27	1	352.27
$X_2 X_1$	33.17	1	33.17
$X_3 X_1, X_2$	11.54	1	11.54
Error	98.41	16	6.15
Total	495.39	19	

Body Fat Example: Testing, $H_0: \beta_3 = 0$ vs. $H_a: \beta_3 \neq 0$

$$F^* = \frac{SSR(X_3|X_1, X_2)}{1} \div \frac{SSE(X_1, X_2, X_3)}{n - 4} = \frac{11.54}{1} \div \frac{98.41}{16} = 1.88$$

$$F^* = 1.88 \leq 8.53 = F(0.99; 1, 16) \Rightarrow \text{conclude } H_0 \text{ (} \alpha = 0.01 \text{)}$$

- X_3 can be dropped from the regression model that already contains X_1, X_2

Test whether a Single $\beta_k = 0$, cont'd

- R Codes For Extra Sum Of Squares With The Body Fat Example:

```
CH07TA01 <- read.csv("/cloud/project/Fall 2020/CH07TA01.csv")
n<-dim(CH07TA01)[[1]]
frm1 <- lm(Y~X1+X2+X3,data=CH07TA01)
frm2 <- lm(Y~X1+X2,data=CH07TA01)
SSE1 <-deviance(frm1)
SSE2 <-deviance(frm2)
F<-((SSE2-SSE1)/1)/(SSE1/(n-4))
F
anova(frm2,frm1)
```

Test whether Several $\beta_k = 0$

- Test whether $\beta_2 X_2$ and $\beta_3 X_3$ can be dropped from the full model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i \quad \text{Full Model}$$

- Alternative

$$H_0: \beta_2 = \beta_3 = 0$$

$$H_a: \text{not both } \beta_2 \text{ and } \beta_3 \text{ equal } 0$$

- When H_0 holds:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \varepsilon_i$$

Reduced Model

Test whether Several $\beta_k = 0$

- Test statistics
 1. $SSE(F) = SSE(X_1, X_2, X_3)$, $df_F = n - 4$
 2. $SSE(R) = SSE(X_1)$, $df_R = n - 2$
 3. The general linear test statistic:

$$\begin{aligned} F^* &= \frac{SSE(R) - SSE(F)}{df_R - df_F} \div \frac{SSE(F)}{df_F} \\ &= \frac{SSE(X_1) - SSE(X_1, X_2, X_3)}{(n - 2) - (n - 4)} \div \frac{SSE(X_1, X_2, X_3)}{n - 4} \\ &= \frac{SSR(X_2, X_3 | X_1)}{2} \div \frac{SSE(X_1, X_2, X_3)}{n - 4} \\ &= \frac{MSR(X_2, X_3 | X_1)}{MSE(X_1, X_2, X_3)} \end{aligned}$$

Test whether a Single $\beta_k = 0$, cont'd

TABLE 7.4
ANOVA Table
with
Decomposition
of *SSR*—Body
Fat Example
with Three
Predictor
Variables.

Source of Variation	<i>SS</i>	<i>df</i>	<i>MS</i>
Regression	396.98	3	132.33
X_1	352.27	1	352.27
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Error	98.41	16	6.15
Total	495.39	19	

Can both X_2 and X_3 be dropped from the full model?

$$F^* = \frac{SSR(X_2, X_3|X_1)}{2} \div MSE(X_1, X_2, X_3) = \frac{33.17 + 11.54}{2} \div 6.15 = 3.63$$

$F^* = 3.63 \sim 3.63 = F(0.99; 2, 16) \Rightarrow$ at the boundary of the decision rule

We may wish to make further analyses before deciding whether X_2 and X_3 should be dropped from the regression model that already contains X_1 .

Summary of Tests Concerning Regression Coefficients

- Test whether **all** $\beta_k = 0$

$$\text{overall } F \text{ test: } F^* = \frac{MSR}{MSE} \sim F(p-1, n-p)$$

- Test whether **a single** $\beta_k = 0$

$$\text{partial } F \text{ test: } F^* = \frac{MSR(X_k | X_1, \dots, X_{k-1}, X_{k+1}, \dots, X_{p-1})}{MSE}$$

$$\sim F(1, n-p)$$

$$\Leftrightarrow t^* = \frac{b_k}{s\{b_k\}}$$

Summary of Tests Concerning Regression Coefficients, cont'd

- Test whether **some** $\beta_k = 0$

$$H_0 : \beta_q = \beta_{q+1} = \cdots = \beta_{p-1} = 0$$

$$\text{partial } F \text{ test: } F^* = \frac{MSR(X_q, \dots, X_{p-1} | X_1, \dots, X_{q-1})}{MSE} \\ \sim F(p - q, n - p)$$

Summary of Tests Concerning Regression Coefficients, cont'd

- When tests about regression coefficients are desired that do not involve testing whether one or several β_k equal zero, extra sums of squares cannot be used and the general linear test approach requires separate fittings of the full and reduced models.

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i \quad \text{Full Model}$$

- Wish to test: $H_o: \beta_1 = \beta_2$ vs. $H_a: \beta_1 \neq \beta_2$

$$Y_i = \beta_0 + \beta_c(X_{i1} + X_{i2}) + \beta_3 X_{i3} + \varepsilon_i \quad \text{Reduced Model}$$

- Wish to test: $H_o: \beta_1 = 3, \beta_3 = 5$ vs. $H_a: \text{not both equalities in } H_o \text{ holds}$
- Under H_o , $\beta_1 X_1$ and $\beta_3 X_3$ are known constants

$$Y_i - 3X_{i1} - 5X_{i3} = \beta_0 + \beta_2 X_{i2} + \varepsilon_i \quad \text{Reduced Model}$$